

Copy-Paste Augmentation Report for MinneApple and Global Wheat Head

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1 Objective

This report aims to show my analysis and thinking about why detection performance decreases when using copy-paste augmentation.

2 Recap

- **Scoring system:** the score is meant to measure how hard each object and image is for the detector. For an object g , we use low-confidence predictions and compute

Let $G(I)$ be the set of ground-truth objects in image I , and let $\mathcal{P}$ be the set of model predictions. For a ground-truth object g denote by

$$\mathcal{P}(g; \tau) = \{ p \in \mathcal{P} : \text{IoU}(p, g) \geq \tau \}$$

the subset of predictions whose intersection-over-union with g exceeds the IoU threshold $\tau \in [0, 1]$. Each prediction p has a confidence $\text{conf}(p) \in [0, 1]$ and an IoU value $\text{IoU}(p, g) \in [0, 1]$.

For $\alpha, \beta \geq 0$ we define the object difficulty score

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g; \tau)} [\alpha(1 - \text{conf}(p)) + \beta(1 - \text{IoU}(p, g))] , & \mathcal{P}(g; \tau) \neq \emptyset, \\ \alpha + \beta, & \mathcal{P}(g; \tau) = \emptyset. \end{cases}$$

This definition has the following properties:

- Lower confidence or lower IoU increases the score (more difficult).
- The upper bound for $S_{\text{obj}}(g)$ is $\alpha + \beta$ (occurs when no matching prediction exists).

The image-level difficulty score aggregates object difficulties and penalizes false positives.

With aggregation weights $w_1, w_2 \geq 0$, we define:

$$S_{\text{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\text{obj}}(g) + w_2 \cdot \text{FP rate}(I; \gamma, \tau).$$

Interpretation: higher $S_{\text{img}}(I)$ indicates that the image contains harder-to-detect objects and/or a larger proportion of false positives.

- **Copy-paste pipeline:** we use this because we believe increasing the appearance of objects and backgrounds can improve performance in agricultural detection problems.

1. Select a background image.

2. Select one or more objects to paste.
3. Determine how many objects to paste (2-8 objects).
4. Apply geometry transformations (scaling and rotation) on selected objects.
5. Paste objects and generate new labels.

3 Experiments

We ran two experiments to compare detection results under baseline, random copy-paste, and score-guided copy-paste.

The same pipeline was used across seeds 123, 456, and 789: dataset ratio 0.3, 2–5 pasted objects per image, scale jitter within $\pm 20\%$, rotation within $\pm 15^\circ$, overlap avoidance enabled, and only the selection policy changed between random and score-guided sampling.

3.1 MinneApple results

Table 1: MinneApple: 3-seed mean COCO AP on the original test set

Condition	AP	AP50	AP75	AP _S	AP _M	AP _L
Baseline	0.439	0.771	0.452	0.300	0.592	0.645
Random CP	0.415	0.741	0.427	0.279	0.563	0.669
Score CP	0.428	0.759	0.440	0.286	0.581	0.637

- MinneApple is the only case where score-guided copy-paste is clearly better than random copy-paste, but the improvement is still not enough to recover the baseline.
- Score CP improves AP, AP50, AP75, and especially AP_S and AP_M relative to Random CP, indicating the score pushes selection toward smaller and medium objects.
- Both augmentation variants still reduce performance compared with the baseline; the gain is only relative, not absolute.

3.2 Global Wheat Head results

Table 2: Global Wheat Head: 3-seed mean COCO AP on the original test set

Condition	AP	AP50	AP75	AP _S	AP _M	AP _L
Baseline	0.507	0.922	0.491	0.136	0.501	0.550
Random CP	0.486	0.905	0.456	0.118	0.480	0.529
Score CP	0.484	0.905	0.455	0.098	0.477	0.531

- Global Wheat Head shows the same failure pattern, with even less separation between random and score-guided selection.
- Both copy-paste variants remain below baseline on every main metric; Score CP is only marginally different from Random CP.
- The gap is too small to support a meaningful claim of improvement; this appears to be augmentation noise rather than a useful intervention.

4 Analysis and Hypotheses

4.1 MinneApple

- **Observation:** The augmented examples still look somewhat artificial, even though the pasted objects are visually plausible (see Appendix Figures 1 and 3).
- **Analysis:** The feature-space and object-paste views suggest that the augmentation is changing the training distribution without clearly moving it toward the test distribution (see Appendix Figures 2 and 3).
- **Hypothesis:** The performance drop comes from pasted-object noise and from augmented images that are not close enough to the test distribution.

4.2 Global Wheat Head

- **Observation:** Score-guided sampling concentrates pasted objects in a narrower portion of the source domains, especially the arvalis domains (see Appendix Figure 7).
- **Analysis:** This concentration does not improve detection, and the feature-space plus AP views still indicate a mismatch with test performance (see Appendix Figures 5 and 6).
- **Hypothesis:** GWHD is dominated by domain shift, so copy-paste alone is unlikely to solve the real problem.

5 Conclusion

- On MinneApple, score-guided copy-paste is better than random copy-paste, especially for small and medium objects, but both still underperform the baseline.
- On Global Wheat Head, random and score-guided sampling are almost the same, and both remain below baseline. Overall, the evidence suggests that copy-paste adds noise and does not reliably move the training data toward the test distribution. For GWHD, the failure mode looks too much like domain shift to be solved by copy-paste alone.

A Appendix: Additional Figures and Details

A.1 Configuration Tables

Table 3: Training configuration (MinneApple vs Global Wheat Head)

Parameter		MinneApple
Epochs		200
Batch size		16
Input size		1024
Learning rate		0.01
Early-stopping patience		10
Seed		123
Pretrained weights	/home/khanh/Projects/DifficultyAgri/finetuned.baseline.model/yolov8s.pt	/hom



Figure 1: MinneApple: augmented vs. original examples.

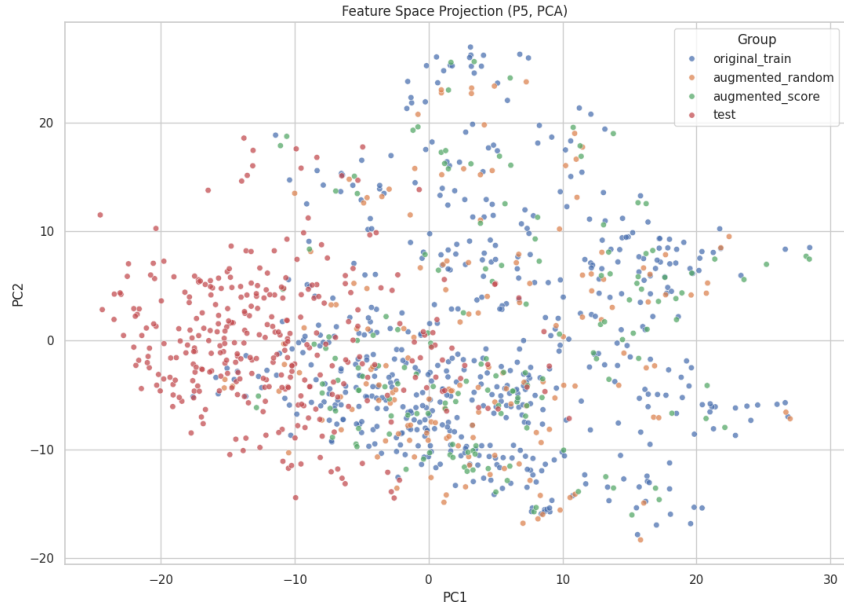


Figure 2: MinneApple: feature-space projection.

Table 4: Scoring configuration

Parameter	Value
Scorer type	<code>min_scorer</code>
Object weight α	0.5
IoU penalty weight β	0.5
IoU threshold τ	0.5
Weight mode	<code>balance_correlation</code>
Selected object weight	0.9
False-positive weight	0.1

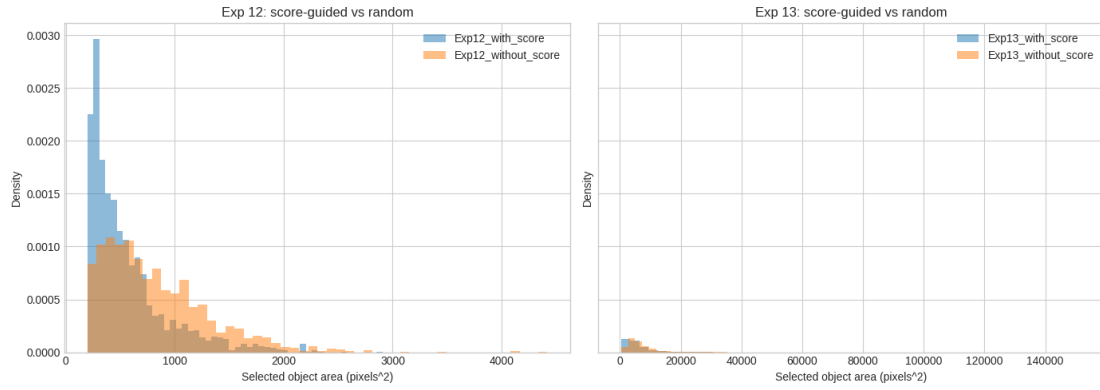


Figure 3: MinneApple: object-paste analysis.

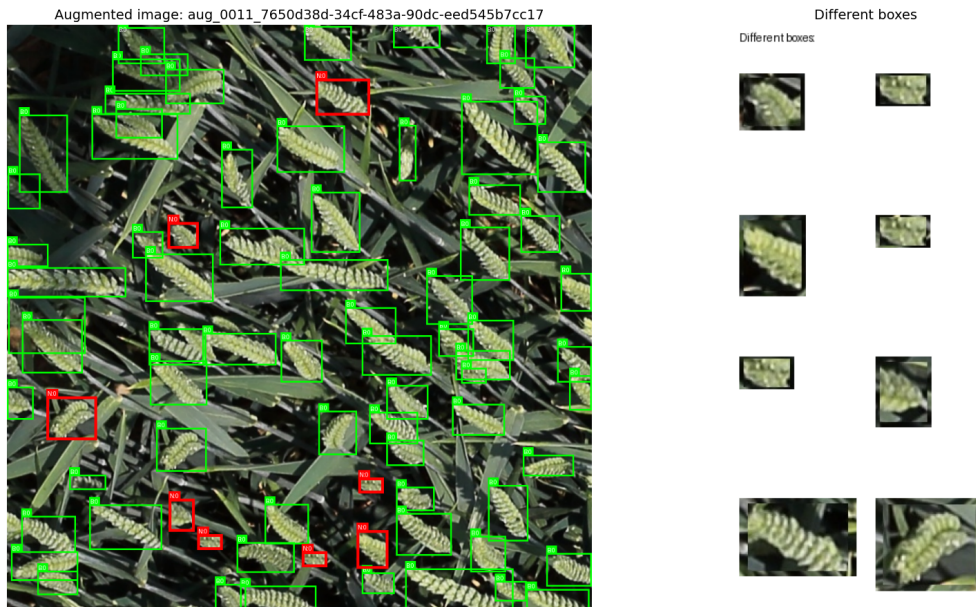


Figure 4: Global Wheat Head: augmented vs. original examples.

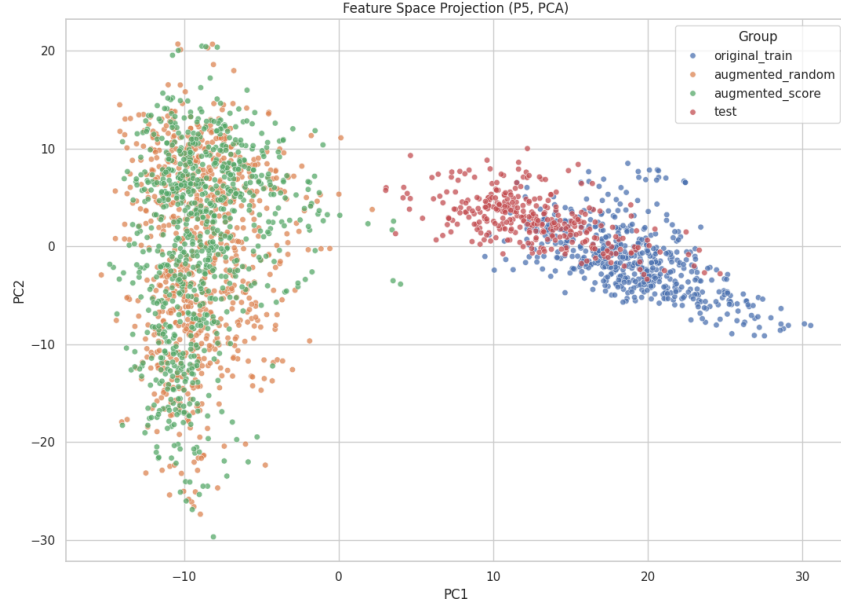


Figure 5: Global Wheat Head: feature-space projection.

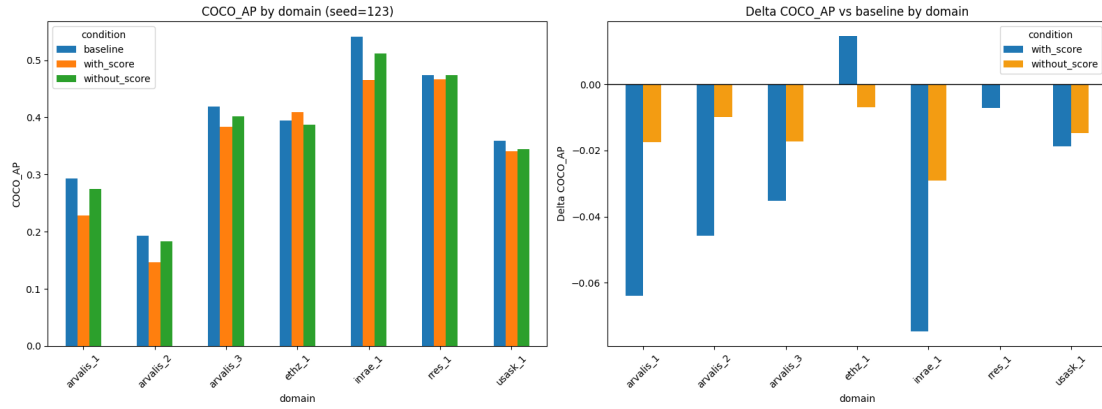


Figure 6: Global Wheat Head: AP comparison result.

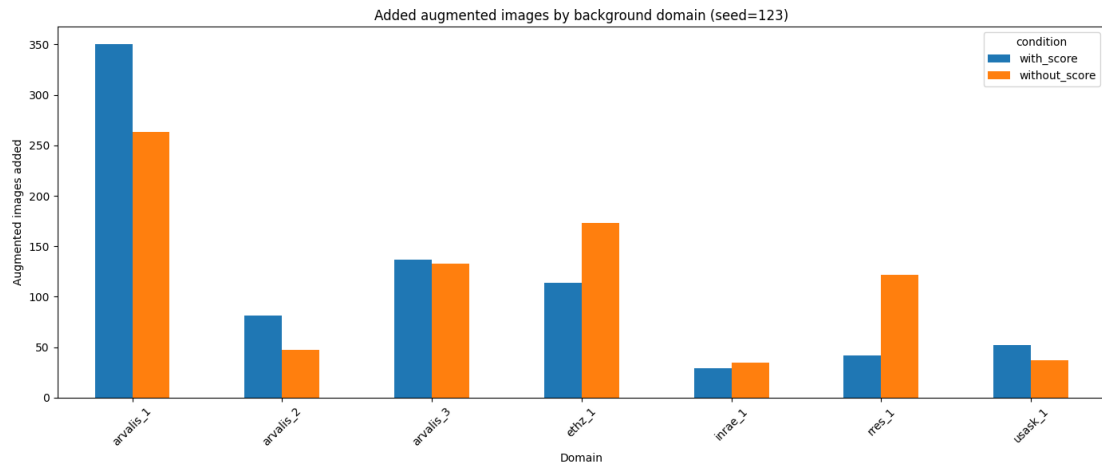


Figure 7: Global Wheat Head: added augmentation distribution.

Table 5: Augmentation / Copy-Paste configuration (MinneApple vs Global Wheat Head)

Parameter	MinneApple	Global Wheat Head
Use score guidance	true	true
Reverse guidance	false	false
Same-image only	true	true
Score weight function	exponential	exponential
Score α	2.0	2.0
Dataset ratio	0.3	0.3
Min objects/image	2	2
Max objects/image	8	8
Selection seed	123	123
Max image reuse	2	3
Max object reuse	3	3
Min object area (px ²)	200	200
Scale range	[0.9, 1.1]	[0.8, 1.2]
Rotation max (deg)	0	0
Blending method	none	none
Avoid overlap	true	true
Placement margin (px)	5	5
Random placement attempts	12	12
Use jiggle placement	true	true